

PLPP Assignment 2

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Git repo link: <https://github.com/VY-Assignments/text-editor-matvii-kovalyshyn>

Program explanation

- New commands overview: which command numbers map to Delete, Undo, Redo, Cut, Paste, Copy, Insert-with-replacement

9 - Delete, 10 - Copy, 11 - Paste, 12 - Cut, 13 - Undo, 14 - Redo, 7 - Insert-with-replacement

- Undo/Redo design: how you chose to snapshot state, why, and what the depth limit means in practice

I chose to save only the changes in a particular row without storing all the data and using extra memory. I also store type of changes and then call reversed actions in Undo function. Redo works the same way.

- Clipboard design: how the clipboard buffer is allocated and managed (can be a RAM internals drawing)

All big and global data is stored in heap using malloc. Text array is stored in the heap during all the time the program works, while copied text is stored in clipboard of this heap and only until it is replaced by another copied text.

- Relationship between commands: explain that Cut = Copy + Delete (or why your implementation differs), and how Paste reuses the same path as Insert

Cut is just calls of Copy and Delete functions because it works exactly the same way. Paste uses Insert function but with copied text to insert.

- What is NOT tracked by Undo/Redo and why

File operations, visual functions (print, clear), search functions and similar functions which do not affect the actual text array. There is no need to undo/redo these actions.

Memory layout diagram

- Show all three heap regions: text buffer, history stack (array of snapshots or command structs), clipboard buffer — with pointers between them

Text Buffer:

```
struct Array textArray
struct Row* rows
int rowCount
int currentRow
```

Clipboard Buffer:

```
struct Array textArray
char* copied
```

History Stack:

```
struct Array textArray
struct HistoryAction* undoLastAction
struct HistoryAction* redoLastAction
```

Test plan — one screenshot per scenario

Base flow:

- Delete → print (matches assignment example)

```
Choose the command: 1
Enter text to append: hello world

Supported commands.
1. Append to end.
2. New line.
3. Save to file.
4. Load from file.
5. Print to console.
6. Insert.
7. Insert with replacement.
8. Search.
9. Delete.
10. Copy.
11. Paste.
12. Cut.
13. Undo.
14. Redo.
0. Exit.

Choose the command: 9
Choose line, index and number of symbols: 0 6 5

Supported commands.
1. Append to end.
2. New line.
3. Save to file.
4. Load from file.
5. Print to console.
6. Insert.
7. Insert with replacement.
8. Search.
9. Delete.
10. Copy.
11. Paste.
12. Cut.
13. Undo.
14. Redo.
0. Exit.

Choose the command: 5

Text:
hello
```

- Undo × 3 → print after each (shows depth)

```
Choose the command: 13

Supported commands.
1. Append to end.
2. New line.
3. Save to file.
4. Load from file.
5. Print to console.
6. Insert.
7. Insert with replacement.
8. Search.
9. Delete.
10. Copy.
11. Paste.
12. Cut.
13. Undo.
14. Redo.
0. Exit.

Choose the command: 5

Text:
hello world

Supported commands.
1. Append to end.
2. New line.
3. Save to file.
4. Load from file.
5. Print to console.
6. Insert.
7. Insert with replacement.
8. Search.
9. Delete.
10. Copy.
11. Paste.
12. Cut.
13. Undo.
14. Redo.
0. Exit.

Choose the command: 13

Choose the command: 13

Supported commands.
1. Append to end.
2. New line.
3. Save to file.
4. Load from file.
5. Print to console.
6. Insert.
7. Insert with replacement.
8. Search.
9. Delete.
10. Copy.
11. Paste.
12. Cut.
13. Undo.
14. Redo.
0. Exit.

Choose the command: 5

Text:
```

- Redo × 3 → print after each (shows depth)

Choose the command: 14 Supported commands. 1. Append to end. 2. New line. 3. Save to file. 4. Load from file. 5. Print to console. 6. Insert. 7. Insert with replacement. 8. Search. 9. Delete. 10. Copy. 11. Paste. 12. Cut. 13. Undo. 14. Redo. 0. Exit.	Choose the command: 14 Supported commands. 1. Append to end. 2. New line. 3. Save to file. 4. Load from file. 5. Print to console. 6. Insert. 7. Insert with replacement. 8. Search. 9. Delete. 10. Copy. 11. Paste. 12. Cut. 13. Undo. 14. Redo. 0. Exit.
Choose the command: 5 Text: hello world Supported commands. 1. Append to end. 2. New line. 3. Save to file. 4. Load from file. 5. Print to console. 6. Insert. 7. Insert with replacement. 8. Search. 9. Delete. 10. Copy. 11. Paste. 12. Cut. 13. Undo. 14. Redo. 0. Exit.	Choose the command: 5 Text: hello Supported commands. 1. Append to end. 2. New line. 3. Save to file. 4. Load from file. 5. Print to console. 6. Insert. 7. Insert with replacement. 8. Search. 9. Delete. 10. Copy. 11. Paste. 12. Cut. 13. Undo. 14. Redo. 0. Exit.
Choose the command: 14	Choose the command: 14

- Cut → print (text removed); Paste → print (text restored at new position)

Choose the command: 12
Choose line, index and number of symbols: 0 6 5

Supported commands.

1. Append to end.
2. New line.
3. Save to file.
4. Load from file.
5. Print to console.
6. Insert.
7. Insert with replacement.
8. Search.
9. Delete.
10. Copy.
11. Paste.
12. Cut.
13. Undo.
14. Redo.
0. Exit.

Choose the command: 5

Text:
hello

Supported commands.

1. Append to end.
2. New line.
3. Save to file.
4. Load from file.
5. Print to console.
6. Insert.
7. Insert with replacement.
8. Search.
9. Delete.
10. Copy.
11. Paste.
12. Cut.
13. Undo.
14. Redo.
0. Exit.

Choose the command: 11
Choose line, index: 0 0

Choose the command: 11
Choose line, index: 0 0

Supported commands.

1. Append to end.
2. New line.
3. Save to file.
4. Load from file.
5. Print to console.
6. Insert.
7. Insert with replacement.
8. Search.
9. Delete.
10. Copy.
11. Paste.
12. Cut.
13. Undo.
14. Redo.
0. Exit.

Choose the command: 5

Text:
worldhello

- Copy → Paste → print (original unchanged, duplicate appears)

```
Choose the command: 10
Choose line, index and number of symbols: 0 5 5
```

Supported commands.

1. Append to end.
2. New line.
3. Save to file.
4. Load from file.
5. Print to console.
6. Insert.
7. Insert with replacement.
8. Search.
9. Delete.
10. Copy.
11. Paste.
12. Cut.
13. Undo.
14. Redo.
0. Exit.

```
Choose the command: 11
Choose line, index: 0 0
```

Supported commands.

1. Append to end.
2. New line.
3. Save to file.
4. Load from file.
5. Print to console.
6. Insert.
7. Insert with replacement.
8. Search.
9. Delete.
10. Copy.
11. Paste.
12. Cut.
13. Undo.
14. Redo.
0. Exit.

```
Choose the command: 5
```

```
Text:
helloworldhello
```

- Insert-with-replacement → print (matches assignment example)

```
Choose the command: 7
Choose line and index: 0 6
Enter text to insert: some text
```

Supported commands.

1. Append to end.
2. New line.
3. Save to file.
4. Load from file.
5. Print to console.
6. Insert.
7. Insert with replacement.
8. Search.
9. Delete.
10. Copy.
11. Paste.
12. Cut.
13. Undo.
14. Redo.
0. Exit.

```
Choose the command: 5
```

```
Text:
hellow some text
```

- Undo a Cut → print (text restored); confirms Undo works for clipboard ops

Choose the command: 12
Choose line, index and number of symbols: 0 6 5

Supported commands.

1. Append to end.
2. New line.
3. Save to file.
4. Load from file.
5. Print to console.
6. Insert.
7. Insert with replacement.
8. Search.
9. Delete.
10. Copy.
11. Paste.
12. Cut.
13. Undo.
14. Redo.
0. Exit.

Choose the command: 5

Text:
hello

Supported commands.

1. Append to end.
2. New line.
3. Save to file.
4. Load from file.
5. Print to console.
6. Insert.
7. Insert with replacement.
8. Search.
9. Delete.
10. Copy.
11. Paste.
12. Cut.
13. Undo.
14. Redo.
0. Exit.

```
Choose the command: 13
```

```
Supported commands.
```

1. Append to end.
2. New line.
3. Save to file.
4. Load from file.
5. Print to console.
6. Insert.
7. Insert with replacement.
8. Search.
9. Delete.
10. Copy.
11. Paste.
12. Cut.
13. Undo.
14. Redo.
0. Exit.

```
Choose the command: 5
```

```
Text:
```

```
hello world
```

Edge cases:

- All new commands on empty buffer: no crash

```
Choose the command: 7
Choose line and index: 0 0
Enter text to insert:
Enter text to insert:
Invalid line or index.
```

```
Choose the command: 9
Choose line, index and number of symbols:
0 0 0
Invalid line or index.
```

```
Choose the command: 10
Choose line, index and number of symbols: 0 0 0
Invalid line or index.
```

```
Choose the command: 11
Choose line, index: 0 0
Invalid line or index.
```

```
Choose the command: 12
Choose line, index and number of symbols: 0 0 0
Invalid line or index.
```

```
Choose the command: 13
```

```
Supported commands.
```

1. Append to end.
2. New line.
3. Save to file.
4. Load from file.
5. Print to console.
6. Insert.
7. Insert with replacement.
8. Search.
9. Delete.
10. Copy.
11. Paste.
12. Cut.
13. Undo.
14. Redo.
0. Exit.

```
Choose the command: 14
```

```
Supported commands.
```

- Undo at bottom of history / Redo with nothing to redo: clear message, no crash

```
Choose the command: 13
```

```
Supported commands.
```

1. Append to end.
2. New line.
3. Save to file.
4. Load from file.
5. Print to console.
6. Insert.
7. Insert with replacement.
8. Search.
9. Delete.
10. Copy.
11. Paste.
12. Cut.
13. Undo.
14. Redo.
0. Exit.

```
Choose the command: 14
```

```
Supported commands.
```

- Paste with empty clipboard: error or no-op with message

```
Choose the command: 11
Choose line, index: 0 0
Invalid line or index.
```

- Delete/Cut with out-of-range line or index: explicit error, buffer unchanged

```
Choose the command: 9
Choose line, index and number of symbols: 100 100 100
Invalid input.
Supported commands.
```

```
Choose the command: 12
Choose line, index and number of symbols: 100 100 100
Invalid input.
```

Git history screenshot (git log / github / gitk)

Commits on Jun 5, 2026
insert with replacement  kkovmat committed 5 days ago 9636d97  
delete function  kkovmat committed 5 days ago 55c258d  
some fixes  kkovmat committed 5 days ago a3936d6  
Commits on May 27, 2026
final commit: some bug fixing and cleaning up code  kkovmat committed 2 weeks ago 2ea630a  
Commits on May 25, 2026
insert done  kkovmat committed 2 weeks ago f91fc58  
file save and load  kkovmat committed 2 weeks ago fc2dc68  
search completed  kkovmat committed 2 weeks ago 5ffa8e9  
Commits on May 24, 2026
append, new line, print done, search started  kkovmat committed 2 weeks ago 5646f50  
Commits on May 23, 2026
add to end completed  kkovmat committed 3 weeks ago 02574a4  
array fixed  kkovmat committed 3 weeks ago e3f2057  
ui and double array start  kkovmat committed 3 weeks ago 76b6414  
Commits on May 21, 2026
first commit  kkovmat committed 3 weeks ago 080cb99  

Commits	
 main	 All users  All time
Commits on Jun 10, 2026	
minor fixes  kkovmat committed 9 minutes ago 47ed716  	
this is the end (i hope)  kkovmat committed 12 hours ago b7ddb03  	
this is the end (i hope)  kkovmat committed 12 hours ago 3b0978f  	
Commits on Jun 9, 2026	
delete, cut undo (it works!)  kkovmat committed 16 hours ago f12aa7d  	
insert undo fix  kkovmat committed 16 hours ago 4a583ec  	
fixing imaginary error  kkovmat committed 16 hours ago 6ecf736  	
Commits on Jun 8, 2026	
undo started and some fixes  kkovmat committed 2 days ago 2026ee5  	
cut/copy/paste, insert, delete  kkovmat committed 2 days ago b8e9f02  	
Commits on Jun 5, 2026	
insert with replacement  kkovmat committed 5 days ago 9636d97  	
delete function  kkovmat committed 5 days ago 55c258d  	